

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry, ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Energy** submission.

Experience

- 2025 – Present

Sustainable Electrochemical Energy Development (SEED) Center, NTUST
Lab Equipment Procurement & Asset Maintenance Manager

 - Procurement (>NTD 2M):** Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths.
 - TGA testing service:** Ran TGA measurements for graduate students from labs across NTUST — Chemical Engineering (Profs. Yu-Cheng Chiu, Min-Hsin Yeh, Young Ku and Yen-Ling Yang), Mechanical Engineering (Prof. Yu-Lin Kuo) and Materials Science & Engineering (Prof. Shih-Yun Chen).
 - Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
 - Maintained close, responsive contact with vendor engineers (Waters, TA Instruments, Uni-Onward), frequently securing on-site repair within **24 hours** of a fault report — significantly reducing instrument downtime.
- Summer 2022

New Taipei City Dept. of Urban Development
Summer Intern, Housing Development Section

 - Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
- 2021 – 2024

Tunghai University Student Housing / GEPT
Resident Assistant (2 yrs) / Exam Staff (part-time)

 - Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters
Partner: a smart-water-meter battery maker

Screening methodology
Designed full-stage test protocols, experimental workflows and statistical validation models.

Database & SOP
Established the test database and standardized high-throughput screening procedures (SOP).

Advanced analysis
EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Training & tech transfer
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy Journal submission — battery thermal stability study

Co-author Responsible for TGA thermal measurement
Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B)**
Certified
- Indoor Air Quality Maintenance Management Specialist**
Certified
- Waste Disposal Technician (Class A)**
Certified
- Occupational Safety and Health Management Personnel (Level B)**
2026 / 11 · mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)**
2027 / 03 · exam scheduled

Education

- 2025 – 2027 (expected)

National Taiwan University of Science and Technology
M.S. in Chemical Engineering
- 2020 – 2024

Tunghai University
B.S. in Chemical & Materials Engineering

Skills & Instruments

- Electrochemistry & batteries**

 - Impedance spectroscopy (EIS)
 - Battery aging analysis
 - Reliability analysis
 - Low-temp electrolytes
- Characterization**

 - Thermal analysis (TGA / DSC)
 - Synchrotron (TLS20A / TPS27A / 45A)
 - Rheology (ARES-G3)
 - Viscometer & densimeter (DMA5002)
- Process & operations**

 - Lithography (nanoimprint / self-assembly)
 - Equipment procurement & maintenance

- Software**

 - Aspen Plus V14
 - Aspen Dynamics
 - Microsoft Word
 - Microsoft Excel
 - Microsoft PowerPoint
 - OriginPro
 - TA Universal Analysis
 - TA TRIOS

Conferences & Training

- Presenter

2026 TwICHe Annual Meeting

Invited training

TA Waters User Seminar
Hands-on instrument operation